Lin Li

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Web: treelli.github.io | **Google Scholar**: dxP6Y_oAAAAJ | **Github**: TreeLLi

Research Experience

Postdoctoral Researcher, Department of Computer Science, University of Oxford, UK — 2025.03 – Present
Advisor: Prof. Yarin Gal (Oxford Applied and Theoretical Machine Learning Group)
Administrative activities: organise group activities like research updates, tea talks; lead grant application and manage the fund; co-supervise MSc and PhD students; support PhD and Postdoc admission like CV screening and interview; represent the group in outreach events.

Research Intern, Robotics X Lab, Tencent, Shenzhen, China — 2021 – 22
Mentor: Prof. Lipeng Chen.
Project: enabling robots to learn throwing and catching from human demonstrations.

Teaching / Supervisory Experience

Mentor, Algoverse — 2026
Program: AI Research Program (mentoring 4 undergraduate students on AI research)

Guest Lecturer, Department of Computer Science, University of Oxford, Oxford, UK — 2025
Course: Uncertainty in Deep Learning (MSc, 20-30 students)

Tutor, Lady Margaret Hall, University of Oxford, Oxford, UK — 2025
Course (Summer School): Computer Vision (2 classes, 10-15 international students per class)

Teaching Assistant, Oxford and KCL — 2021 / 25
Courses: Uncertainty in Deep Learning (MSc), Machine Learning and Pattern Recognition (MSc), Introduction to Artificial Intelligence (Undergraduate)

Education

PhD in Machine Learning, Department of Informatics, King's College London, UK — 2019 – 24
Advisor: Prof. Michael Spratling. Thesis Committee: Prof. George D. Magoulas and Dr. Adel Bibi.
Thesis: *Towards Robust Visual Classification through Adversarial Training*

MSc in Computing Science, Department of Computing, Imperial College London, UK — 2017 – 18
Advisor: Prof. Wayne Luk. Grade: Distinction (First Class Honour)

BM in Financial Management, School of Management, Xiamen University, China — 2013 – 17
Advisor: Prof. Zheng Qiao. Grade: top 10% of graduating class

Publication (* indicates equal contribution; † indicates the corresponding author if have)

Overview: 13 peer-reviewed publications (8 first-authored), 783 citations (Google Scholar).

Core AI:

1. **Lin Li**, Georgia Channing, Suhaas M Bhat, Gabriel Davis Jones, Yarin Gal, "Building Reliable Long-Form Generation via Step-Wise Hallucination Rejection Sampling", ICLR Workshop Agentic AI in the Wild: From Hallucinations to Reliable Autonomy, 2026

2. **Lin Li**†, Michael Spratling, "Robust Shortcut and Disordered Robustness: Improving Adversarial Training Through Adaptive Smoothing", the *Pattern Recognition (**PR**)*, 2025
3. **Lin Li**, Yifei Wang, Chawin Sitawarin, Michael Spratling, "OODRobustBench: a Benchmark and Large-Scale Analysis of Adversarial Robustness under Distribution Shift", the *International Conference on Machine Learning (**ICML**),* 2024
4. **Lin Li***, Haoyan Guan*, Jianing Qiu, Michael Spratling, "One Prompt Word is Enough to Boost Adversarial Robustness for Pre-trained Vision-Language Models", the *IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR**)*, 2024
5. **Lin Li**†, Jianing Qiu, Michael Spratling, "AROID: Improving Adversarial Robustness Through Online Instance-Wise Data Augmentation", the *International Journal of Computer Vision (**IJCV**)*, 2024
6. **Lin Li**, Michael Spratling, "Data augmentation alone can improve adversarial training", the *International Conference on Learning Representations (**ICLR**)*, 2023
7. **Lin Li**, Michael Spratling†, "Understanding and combating robust overfitting via input loss landscape analysis and regularization", the *Pattern Recognition (**PR**)*, 2023

Applied AI:

1. Boqi Chen, Xudong Liu, Jiachuan Peng, Marianne Frey-Marti, Bang Zheng, Kyle Lam, **Lin Li**, Jianing Qiu, "MEDSYN: Benchmarking Multi-EviDence SYNthesis in Complex Clinical Cases for Multimodal Large Language Models", *Findings of the Association for Computational Linguistics: **ACL**,* 2026
2. Jahan C Penny-Dimri, Magdalena Bachmann, William R Cooke, Sam Mathewlynn, Samuel Dockree, John Tolladay, Jannik Kossen, **Lin Li**, Yarin Gal, Gabriel Davis Jones, "Measuring large language model uncertainty in women's health using semantic entropy and perplexity: a comparative study", the *Lancet Obstetrics, Gynaecology, & Women's Health (**Lancet OGWH**)*, 2025
3. Ravi Shankar, Sheng Wong, **Lin Li**, Magdalena Bachmann, Alex Silverthorne, Beth Albert, Gabriel Davis Jones, "Energy Landscapes Enable Reliable Abstention in Retrieval-Augmented Large Language Models for Healthcare", *NeurIPS Workshop GenAI4Health*, 2025
4. Lipeng Chen*†, Jianing Qiu*, **Lin Li***, Xi Luo, Guoyi Chi, Yu Zheng, "Advancing Robots with Greater Dynamic Dexterity: A Large-Scale Multi-View and Multi-Modal Dataset of Human-Human Throw&Catch of Arbitrary Objects", the *International Journal of Robotics Research (**IJRR**)*, 2024
5. Jianing Qiu*, Jian Wu*, Hao Wei*, and Peilun Shi, Minqing Zhang, Yunyun Sun, **Lin Li**, … (+34 authors), Wu Yuan†, "Development and Validation of a Multimodal Multitask Vision Foundation Model for Generalist Ophthalmic Artificial Intelligence", the *New England Journal of Medicine Artificial Intelligence (**NEJM AI**)*, 2024
6. Jianing Qiu, **Lin Li**, Jiankai Sun, and Jiachuan Peng, Peilun Shi, Ruiyang Zhang, Yinzhao Dong, Kyle Lam, Frank P.-W. Lo, Bo Xiao, Wu Yuan, Dong Xu†, Benny Lo†, "Large AI Models in Health Informatics: Applications, Challenges, and the Future", the *IEEE Journal of Biomedical and Health Informatics (**JBHI**)*, 2023

Preprint / Under Review:

1. **Lin Li**, Qi Zhang, Jianing Qiu, Yarin Gal, "AI-Driven Peer Review Could Destroy the Scientific Publication System", *in submission*
2. Hengyang Yao*, **Lin Li***, Ke Sun, Jianing Qiu, Huiping Chen, "Towards Robust Protective Perturbation against DeepFake Face Swapping", *in submission*
3. Xingrui Gu, Chuyi Jiang, Luckeciano Carvalho Melo, Mengyue Yang, **Lin Li**†, "Laplacian Flows for Policy Learning", *in submission*
4. Xinyu Li, Ronghui Mu, **Lin Li**, Tianjin Huang, Gaojie Jin, "OTora: A Unified Red Teaming Framework for Reasoning-Level Denial-of-Service in LLM Agents", *in submission*
5. Linyuan Li, Anujit Saha, **Lin Li**, Boyuan Li, Mengxian He, Jianing Qiu, Wu Yuan, "Artificial Intelligence for Biomedical Video Generation", *in submission*
6. Jianing Qiu, **Lin Li**, Jiankai Sun, Hao Wei, Zhe Xu, Kyle Lam, Wu Yuan, "Emerging Cyber Attack Risks of Medical AI Agents", *in submission*

Grants & Awards

Advancing Large AI Models: Integration of New Data Modalities and Expansion of Capabilities 2025-29
Amount: €24.98M. Funder: Horizon Europe
Role: Proposal contributor (one of the seven work packages), and Research project leader. PI: Prof. Yarin Gal.

Technical AI Safety Research 2026-27
Amount: $371K. Funder: Coefficient Giving (Open Philanthropy)
Role: Co-Principal Investigator (lead the grant application and the project).

Proactive Defense against Deepfake 2025
Amount: £20K. Funder: IDAI Pump Prime, University of Birmingham
Role: Co-Investigator. PI: Prof. Huiping Chen.

Scholarships & Honours

OpenAI API Access (Credits $500) 2025
Nomination of USERN Prize 2025
KCL King's-China Scholarship (£109.8K) 2019-24
KCL PGR Research Support (£2.5K) 2023
Xiamen University First Class Scholarship (¥7K) 2016
3rd prize winner in Jinyuan Creativity and Startup Contest 2016
Xiamen University Excellent Academic Performance Scholarship 2015
3rd prize winner in ChinaNet Dream Accelerator Programming Contest 2015

Press

1. **Süddeutsche Zeitung (KZ)**, major national newspapers, German 2025
   Article: Are Language Models Becoming Even More Unreliable?
2. **South China Morning Post (SCMP)**, major national newspapers, Hong Kong, China 2025
   Article: DeepSeek's AI in hospitals is 'too fast, too soon', Chinese medical researchers warn
3. **South China Morning Post (SCMP)**, major national newspapers, Hong Kong, China 2024
   Article: A Hong Kong AI model proves more accurate than doctors in diagnosing eye conditions

Talks

1. **Deep Dive: AI Robustness and Safety**, ELLIS Oxford, UK 2026
   Talk: From Robustness to Safety: Building Trustworthy Systems for High-Stakes Applications.
2. **AI Masterpiece Talk**, Outstanding Young Scholars Society, UK 2025
   Talk: Towards Robust and Safe AI: An Adversarial Approach.
3. **AI Agents for Surgery Workshop**, 17th Hamlyn Symposium on Medical Robotics, UK 2025
   Talk: Hallucination of AI Agents.
4. **CHI Seminar**, Computational Health Informatics (CHI) Lab, University of Oxford, UK 2025
   Talk: Prompting VLMs for accuracy and robustness.
5. **Nanqiang Youth Scholar Forum**, Xiamen University, China 2024
   Talk: Towards Safe Multimodal Foundational Models in Finance.
6. **AI Time Youth PhD Talk**, Tsinghua University, China. 2023
   Talk: Data augmentation can improve adversarial training.
7. **Departmental Research Showcase**, KCL, UK 2023
   Talk: Defending DNNs against adversarial examples.

Academic Service

Reviewer
Conferences: NeurIPS, ICML, ICLR, CVPR, AAAI, ECCV.
Journals: IEEE T-Pattern Analysis and Machine Intelligence (T-PAMI), Pattern Recognition (PR), IEEE T-Information Forensics & Security (T-IFS), IEEE T-Dependable and Secure Computing (T-DSC), IEEE Journal of Biomedical and Health Informatics (J-BHI).

Area Chair
Workshops: Agentic AI for Medicine Workshop@MICCAI2025

Program Committee
Workshops: 2nd MEIS Workshop@CVPR2025, WIHR@ICRA 2024, Workshop on Cooperative Intelligence for Embodied AI@ECCV2024, Agentic AI for Medicine Workshop@MICCAI2025

Academic Referees (refer to their websites for the latest information)

Yarin Gal
Professor of Machine Learning, Department of Computer Science, University of Oxford.
Turing AI Fellow, The Alan Turing Institute, London, UK.
Expert Advisor, UK AI Security Institute (AISI), London, UK.
Tutorial Fellow, Christ Church College, University of Oxford.
Phone: +44 1865 610497, Email: yarin@cs.ox.ac.uk.
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Michael Spratling
Researcher, Department of Behavioural and Cognitive Sciences, University of Luxembourg.
was Reader[1] in Computational Neuroscience and Visual Cognition, Department of Informatics, KCL.
Phone: +352 46 66 44 5722, Email: michael.spratling@uni.lu.
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Adel Bibi
Senior Researcher in ML, Department of Engineering Science, University of Oxford.
Research Fellow at Kellogg College, University of Oxford.
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[1] Michael held this position while serving as Lin's PhD supervisor.